

materials necessary for the antennas, and scaling up the manufacturing. This is one of the technologies that has the potential to significantly improve electricity generation.

12 PHAGE THERAPY

Researchers are finding it hard to develop new antibiotics at the same rate as the bacteria in the wild are developing resistance to them. The problem is aggravated by the indiscriminate, irresponsible and excessive use of antibiotics, which allow bacteria to develop a resistance to treatments quicker. Bacteriophages are viruses that kill bacteria, and were identified as a treatment for bacterial infections back in the 1940s itself. However, at that time, antibiotics emerged as a viable treatment option. Now, researchers are taking a closer look at bacteriophages again. The problem is that the viruses that attack bacteria are highly specific to particular strains of bacteria, which is why researchers are developing potent bacteriophage cocktails. These cocktails include naturally occurring as well as bioengineered phages. Clinical trials are slated to begin in Europe within the next year.

13 PLASMA PROPULSION ENGINE

Electric propulsion has been around for decades, and is common in spacecraft for raising the orbits of satellites deployed into orbit, or for keeping a spacecraft within a particular orbit. The Dawn and Deep Space 1 used ion thrusters for interplanetary travel. Plasma propulsion devices in development use plasma to generate thrust. The Australian National University and the European Space Agency are both involved in developing this new type of engine, which like the ion thrusters, are suitable for interplanetary travel. The Ad Astra Rocket Company is developing an engine known as Variable Specific Impulse Magnetoplasma Rocket (VASIMIR), which will be capable of sending crewed missions to Mars in as little as 39 days. Another potential application of it is for space tugs, to move about cargo and supplies.

14 SCREENLESS DISPLAYS

Screenless displays are any kind of technology that can bypass the use of a screen. Some approaches can directly project the image onto the retina, which significantly improves privacy. Such a projection is visible only to the person operating the machine, and prevents shoulder surfing entirely. Another approach is to bypass the biological eye, and directly provide the information to the optical nerve. This would allow blind people to interface with technology. Rudimentary efforts have managed to convey braille characters to humans, and a video streaming solution has been demonstrated with horseshoe crabs.

15 SEMANTIC WEB

This is a proposed way in which the internet can be made easier to read by machines, opening up new applications. There is a vast amount of information on the internet, which is plagued by uncertainty, inconsistency and general disorganisation. This is made worse with misinformation thrown in. A semantic web would significantly improve the way in which data is retrieved and processed, improving and speeding up decisions. Skeptics claim that the web will never be completely semantic, and even question the usefulness of such a network.

16 VOLUMETRIC DISPLAYS

Holographic displays are a staple of science fiction, but have not been put into production yet. There are some volumetric displays that have been created by labs and startups. One of the biggest roadblocks is the sheer computing power and bandwidth required to support such displays. This is because if you add a dimension to a 1600 x 1200 display, and make it a 1600 x 1200 x 1600 display, you need to have 1600 times the imaging requirements of a 1600 x 1200 display. Now imagine keeping that up at 60 FPS, and you have some kind of an idea of the computing power required. There is also the requirement of these displays to be responsive enough to gestures in near real time to have the displays that Iron Man uses.

17 VOICE COMPUTING

This is something that should be here already. Although Windows 10 has Cortana built in, we do not see too many people using voice computing on a regular basis. While we can imagine it to be incredibly distracting in office, there is no reason why we cannot use it at home. It is possible to use a combination of touch, keyboard, and voice inputs to edit a video on some laptops, but the limitations of each input mechanism must be clearly understood. Unfortunately, we cannot just tell Photoshop to remove the background, add a watermark, and colour correct a photograph just yet. The capabilities of voice assistants can be significantly expanded, neural networks can improve when it comes to processing the speech of humans, and we really need machines to understand Indian accents better. In its current state, our devices are like Professor Cuthbert Calculus, incredibly intelligent but obstinate that they have heard the correct thing even when they have not.

18 WIRELESS POWER

Tesla had a dream of broadcasting power wirelessly to the entire world from a single tower. We would settle for a network of towers, but we really need wireless electricity for powering our devices. With the rollout of 5G and a range of emerging IoT applications, if wireless power transmission is not realised anytime soon, we will need swarms of robots just to go around replacing batteries in all the sensors and smart fixtures in a single smart city. Wearables and smartphones stand to benefit tremendously if the batteries are always charging up wirelessly. We do not even need it to be fast, just around all the time and continuous. Smartphone batteries are turning out to be the biggest pain point of the technology, and are invading an increasing amount of real estate within a device. While technologies for far field wireless energy transmission are being developed, their range is limited, and the efficiency is not that great. About 40 percent of the electricity is lost in the transmission. 